Double GREG estimator

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1 Problem setup

- Let A be a sample of size n selected from the population U of size N. Let π_i be the first order inclusion probability of unit i and π_{ij} the second order inclusion probability of unit i and j.

- Let y denote a variable of interest and we try to estimate the population total $t_y = \sum_{i \in U} y_i$. Let $\mathbf{x}_i$ of size p be the auxiliary variables associated with y. Assume that $\mathbf{x}_i$ is observed throughout the finite population while y_i is observed only in the sample.

- Suppose the following superpopulation model holds

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} (0, \sigma^2), \quad i = 1, \cdots, N \tag{1}$$

- Indeed, the linear regression model can be extended to the general nonlinear model whose basis function can be (penalized) spline functions. (1) can be replaced by

$$y_i = \sum_{j=0}^{M} h_j(\mathbf{x}_i)\beta_j + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d.}{\sim} (0, \sigma^2), \quad i = 1, \cdots, N \tag{2}$$

  where $h_j(\cdot)$ are the corresponding basis functions. In this article, for the sake of simplicity, we assume linear basis functions in (1) only.

- The working model (1) motivates the GREG estimator defined by

$$\hat{t}_y = \sum_{i \in U} \mathbf{x}_i^T \hat{\boldsymbol{\beta}} + \sum_{i \in A} \frac{1}{\pi_i} \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}} \right) \tag{3}$$

  where $\hat{\boldsymbol{\beta}}$ is defined by

$$\hat{\boldsymbol{\beta}} = \left(\sum_{i \in A} \frac{1}{\pi_i} (\mathbf{x}_i - \bar{\mathbf{x}}) (\mathbf{x}_i - \bar{\mathbf{x}})^T \right)^{-1} \left(\sum_{i \in A} \frac{1}{\pi_i} (\mathbf{x}_i - \bar{\mathbf{x}}) y_i \right)$$

(3) enjoys many desirable design properties including asymptotic equivalence to the difference estimator which is defined by replacing the estimated $\hat{\boldsymbol{\beta}}$ by the known $\boldsymbol{\beta}$ in (3). The GREG estimator is also known to be design consistent, asymptotically design unbiased, and asymptotically normally distributed with finite variance. Furthermore, its design variance can consistently be estimated by the linearized variance estimator

$$\hat{V}_p = \sum_{i,j \in A} \frac{\Delta_{ij}}{\pi_{ij}} \frac{y_i - \mathbf{x}_i \hat{\boldsymbol{\beta}}}{\pi_i} \frac{y_j - \mathbf{x}_j \hat{\boldsymbol{\beta}}}{\pi_j} \tag{4}$$

However, when the number of parameters p grows as n increases, the properties listed above no longer hold.

2 Double GREG estimator

- Now, suppose that p is allowed to increase as the sample size increases. In the high-dimension setup, the estimating errors of the regression coefficients $\hat{\boldsymbol{\beta}}$ are no longer negligible and the GREG estimator is no longer asymptotically equivalent to the difference estimator since

$$\hat{t}_{y,GREG} - \hat{t}_{y,diff} = \left(\sum_{i \in U} \mathbf{x}_i - \sum_{i \in A} \frac{1}{\pi_i} \mathbf{x}_i \right) \left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \right) \neq o_p(n^{-1})$$

  Indeed, Ta et al. (2020) recently showed that the GREG estimator is asymptotically equivalent to the difference estimator only if $p/n \to 0$.

- Furthermore, unlike fixed p where the bias ($O(n^{-1})$) of the GREG estimator is negligible compared to the standard error($O(n^{-1/2})$) (Cochran, 1977), the bias of the high dimensional GREG estimator can be significant. The simulation results show that such a tendency manifests itself even stronger when the sampling design is informative (in the next section). One possible way to remove bias is by applying the jackknife method to the GREG estimator, but it does not account for high dimensional variables.

- In this regard, we consider the sample split method. Let A_1 and A_2 be a random partition of A (possibly of equal size). Let $\hat{\boldsymbol{\beta}}_1$ and $\hat{\boldsymbol{\beta}}_2$ be the generalized regression coefficients using A_1 and A_2 respectively. We propose the double GREG estimator defined as follows

$$\hat{t}_y = \sum_{i \in U} \mathbf{x}_i^T \tilde{\boldsymbol{\beta}} + \sum_{i \in A_1} d_i \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_2 \right) + \sum_{i \in A_2} d_i \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 \right) \tag{5}$$

  where $\tilde{\boldsymbol{\beta}}$ is the weighted regression coefficient defined by

$$\tilde{\boldsymbol{\beta}} = \frac{\sum_{i \in A_2} d_i}{\sum_{i \in A} d_i} \hat{\boldsymbol{\beta}}_1 + \frac{\sum_{i \in A_1} d_i}{\sum_{i \in A} d_i} \hat{\boldsymbol{\beta}}_2 \tag{6}$$

- Because of the random split, the resulting double GREG estimator can suffer from large variance. In order to reduce the variance and improve efficiency, we take several random splits from a sample and take the average of all the samples. That is, for each of the random sample $A_1(1), \cdots, A_1(K)$ and $A_2(1), \cdots, A_2(K)$ the final ensemble estimator is defined as

$$\hat{t}_y = \frac{1}{K} \sum_{k=1}^{K} \hat{t}_y(A_1(k), A_2(k)) \tag{7}$$

- Observe that both (5) and (7) are a linear estimator of y. That is, the double estimator can be constructed as a weighted sum of y's whose weights are not dependent on y. The proposed methods can be useful when multivariate y's are used.

- If the design matrix $\mathbf{X}$ has an intercept, (5) is algebraically equivalent to

$$\hat{t}_y = \sum_{i \in U} \mathbf{x}_i^T \tilde{\boldsymbol{\beta}} + \left(\sum_{i \in A_1} d_i \mathbf{x}_i - \sum_{i \in A_2} d_i \mathbf{x}_i \right)^T \left(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_2 \right)$$

- For the variance estimation, we suggest two possible estimators

  - Consider the linearized variance estimation in (4). Instead of $\mathbf{x}_i^T \hat{\boldsymbol{\beta}}$, we replace it by its ensemble estimator

$$\frac{1}{K} \sum_{k=1}^{K} \mathbf{x}_i^T \left(\hat{\boldsymbol{\beta}}_1^{(k)} I\left\{ i \in A_2(k) \right\} + \hat{\boldsymbol{\beta}}_2^{(k)} I\left\{ i \in A_1(k) \right\} \right)$$

  - Recall that the double (or ensemble) estimator defined in (5) is linear in y_i. Writing (5) as

$$\hat{t}_y = \sum_{i \in A} w_i y_i$$

    we can consistently estimate the variance by

$$\sum_{i,j \in A} \frac{\Delta_{ij}}{\pi_{ij}} y_i w_i y_j w_j$$

- Theoretical justification needs to be investigated.

3 Simulation study

Let A be a sample of size $n = 100$ selected from the population U of size $N = 2000$. Let y be a variable of interest and we try to estimate the population total $t_y = \sum_{i \in U} y_i$ and $\mathbf{x}_i$ of size

p be the auxiliary variables associated with y where $x_{ij} \overset{i.i.d.}{\sim} N(2,1)$. Suppose the following superpopulation model holds

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \overset{i.i.d.}{\sim} N(0, \sigma^2), i = 1, \cdots, N$$

where $\sigma^2 = 1$ and $\boldsymbol{\beta} = (1, \cdots, 1, 0, \cdots, 0)$ with $s = 20$ 1s and $p - s = 20$ 0s. We assume the samples are drawn from the informative stratified sampling.

$$Z_j \sim Z_j^* + \varepsilon_j, \quad Z^* \sim N\left(0, \frac{1-r}{r}\right), j \in U, r \in (0,1)$$

The finite population data $\{y_j, x_j, z_j\}_{j \in U}$ are sorted by z_j so that the 500 smallest z values are in stratum one and the next 500 smallest value are in stratum two and so forth. Within each stratum, simple random samples are collected with size

$$n_1 = 15, n_2 = 20, n_3 = 30, n_4 = 35$$

The larger r is, the more informative sampling design we have. We consider the following estimators. Here, we set $r = 0.75$.

- Difference estimator

$$\hat{t}_y = \left(\sum_{i \in U} \mathbf{x}_i\right)^T \boldsymbol{\beta} + \sum_{i \in A} d_i \left(y_i - \mathbf{x}_i^T \boldsymbol{\beta}\right)$$

- GREG estimator

$$\hat{t}_y = \left(\sum_{i \in U} \mathbf{x}_i\right)^T \hat{\boldsymbol{\beta}} + \sum_{i \in A} d_i \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}\right)$$

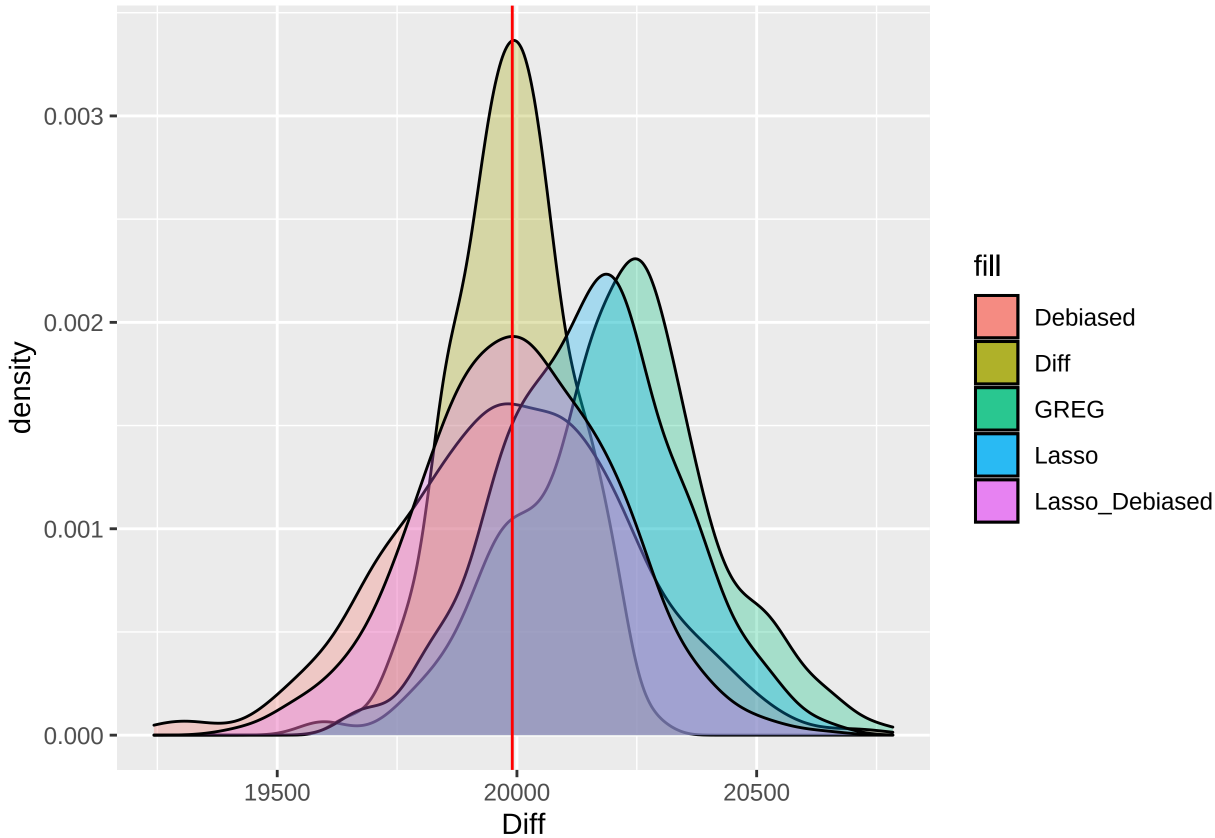
  where $\hat{\boldsymbol{\beta}} = \left(\sum_{i \in A} d_i \mathbf{x}_i \mathbf{x}_i^T\right)^{-1} \left(\sum_{i \in A} d_i \mathbf{x}_i y_i\right)$

- Debiased GREG estimator(Proposed)

$$\hat{t}_y = \left(\sum_{i \in U} \mathbf{x}_i\right)^T \left(\frac{\sum_{i \in A_2} d_i}{\sum_{i \in A} d_i} \hat{\boldsymbol{\beta}}_1 + \frac{\sum_{i \in A_1} d_i}{\sum_{i \in A} d_i} \hat{\boldsymbol{\beta}}_2\right) + \sum_{i \in A_1} d_i \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_2\right) + \sum_{i \in A_2} d_i \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1\right)$$

  where A_1 and A_2 are the subsamples of A with $A_1 \bigcup A_2 = A$, and $\hat{\boldsymbol{\beta}}_1$ and $\hat{\boldsymbol{\beta}}_2$ are the regression coefficients built upon the sample A_1 and sample A_2 respectively.

```
##                        BIAS       SE     RMSE
## HT               14.17026931 344.5636 344.8548
## Diff             -0.08421959 118.2858 118.2858
## GREG            218.84918216 198.7317 295.6167
## Lasso           161.59757722 181.5888 243.0808
## Debiased          3.79573031 240.6009 240.6309
## Lasso_Debiased   12.38923464 198.2491 198.6358
```



References

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